

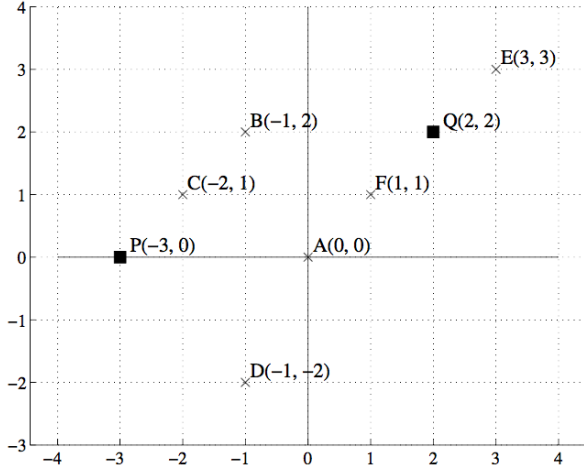
Midterm 2 Part b

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I. QUESTION (15 POINTS)

Suppose we are given the following dataset. Run hierarchical clustering algorithm to cluster the points A, B ... F. The between cluster distance is calculated as the minimum distance between both cluster points, that is, the distance between two clusters C_1 and C_2 is the minimum distance between all pairs of points in C_1 and C_2 .



II. QUESTION (15 POINTS)

If each base-learner is independently and identically distributed and correct with probability $p > 1/2$, what is the probability that a majority vote over L classifiers gives the correct answer? Calculate the probability for $L = 5$ and $p = 3/4$.

III. QUESTION (15 POINTS)

Consider the alternative error function in linear perceptron for regression problem:

$$E = \frac{1}{2}(r - y)^2 + \alpha \sum_i w_i^2 \quad (1)$$

where N represents the number of instances, r represents the actual output, y represents the calculated output, v_i are the weights between the input units and the output unit. Derive the gradient descent update rules for the weights.

IV. QUESTION (15 POINTS)

Let say the values of N data points in 1-dimensional data represented as x . Let sat there are two classes (0, 1) and the classes of the data are stored in array y . Let also assume that the data points in x are already sorted. Either write an algorithm (in pseudocode) to find the best split according to the information gain, or run the algorithm on data:

$x = [1, 3, 5, 7, 11, 15, 19, 23]$

$y = [0, 0, 1, 1, 1, 1, 0, 0]$

`bestSplit(x, y)`

$\log 3 = 1.58$

$\log 5 = 2.32$

$\log 6 = 2.58$

$\log 7 = 2.81$

$\log 4 / 7 = -0.81$

$\log 3 / 7 = -1.23$

$\log 1 / 3 = -1.58$

$\log 2 / 3 = -0.58$

$\log 3 / 5 = -0.74$

$\log 2 / 5 = -1.32$